

Supriya Khadka

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Education

Incoming PhD in Information Technology

Aug 2026 - Aug 2031

George Mason University, USA

Master in Science by Research, AI

Sept 2024 - Sept 2025

Coventry University, United Kingdom

Supported by British Council Women in STEM Scholarship

Bachelors in Computer Engineering

Nov 2018 - Apr 2023

Pulchowk Campus, Tribhuvan University, Nepal

(**Agg. 85.60%**)

Research Experience

Visiting Researcher, CHAI Lab, Northeastern University

Oct 2025 - July 2026

Research Area: Empathy in Conversational Agents

Advisor: [Dr. Smit Desai](#)

- Conducted a systematic review of 89 studies on empathy in conversational agents, synthesizing design, evaluation, and application trends.
- Developed a conceptual framework for empathy in conversational agents spanning interaction contexts, design strategies, and evaluation methods.

Postgraduate Researcher, Coventry University

Sept 2024 - Sept 2025

Project Title: Lowering the Cost of Healthcare by Clinical Coding with LLMs

Advisors: [Dr. Xiaorui Jiang](#), [Prof. Vasile Palade](#)

- Explored the viability of LLMs in clinical coding for evidence extraction, code prediction, and code verification.
- Conducted quality and error analysis on existing clinical coding datasets using LLMs, in collaboration with professional clinical coders.
- Evaluated the performance and cost-effectiveness of different few-shot learning strategies for adapting pre-trained language models (PLMs) and LLMs to the clinical coding tasks.

Undergraduate NLP Research (CARD & NAAMII)

July 2022 - May 2023

Advisors: [Dr. Basanta Joshi](#) (CARD), [Dr. Bishesh Khanal](#) (NAAMII)

- Developed a Nepali Text-to-Speech (TTS) model, leading to a publication and presentation at SIGUL; work included integrating the model into a mobile application at CARD.
- Prepared and publicly released a comprehensive Nepali TTS dataset ([OpenSLR](#)) to support the research.
- Analysed and curated Nepali datasets for Sentiment Analysis, TTS, and Machine Translation at NAAMII, documenting their features and limitations.

Teaching Experience

Undergraduate Research Coordinator, Sunway College Kathmandu Jan 2026 - Jun 2026

- Taught a semester-long course on Basic Research and Study Skills, covering research question formulation, literature review, experimental design, and academic writing, alongside supervising Individual Honours Projects.
- Mentored 18 undergraduate students on individual and group research projects, resulting in six publications at peer-reviewed international workshops and conferences.
- Organized and supervised a weekly paper reading club to build students' research literacy.

Technical Lead and Facilitator, Women Leaders in Technology Sept 2020 - Aug 2024

- Served as a Technical Facilitator for four consecutive batches of a fellowship program aimed at empowering women leaders in technology, after graduating as a fellow.
- Instructed **over 60 fellows** in essential technical subjects, including Git, Basic JavaScript concepts, Asynchronous JavaScript concepts, Express, and MongoDB, during the fellowship bootcamp.
- Promoted to the Technical Lead in the latest batch, responsible for developing the curriculum, training junior facilitators, overseeing all technical sessions and projects, and teaching approximately half of the classes throughout the fellowship.

Industry Experience

Conversational AI Engineer, Diyo.AI May 2023 - Aug 2024

Advisor: Dr. Bishesh Khanal

- Built multilingual chatbots tailored for domains such as sexual and maternal reproductive health, safe abortion, e-governance, and sales.
- Researched and implemented LLMs like *GPT3.5*, *Llama2*, *BLOOM* and *Okapi* to develop and improve conversational models and their language understanding capability for Nepali.
- Curated, refined, and enhanced NLP datasets for low-resource languages to develop language technologies for Nepali and other languages of Nepal, as a continuation of my internship at NAAMII.

Publications & Manuscripts

Peer-Reviewed Papers

Khadka, Supriya, and Sanchari Das. “Privacy-Preserving Compliance on Public Ledgers via Selective Disclosure Authorization Schemes”. *Proceedings of the 23rd International Conference on Security and Cryptography (SECRYPT 2026)*. In Press. SciTeP, 2026.

Khadka, Supriya, and Sanchari Das. “SoK: Understanding the Pedagogical, Ethical, and Privacy Challenges in Adopting XR for Early Childhood Education”. *Proceedings of The Augmented Humans International Conference 2026 (AHs 2026)*. ACM, 2026, <https://doi.org/10.1145/3795011.3795019>.

Khadka, Supriya, and Smit Desai. “Empathy through the Lens of Conversational Agents: A Systematic Review”. *Proceedings of the ACM Conversational User Interfaces Conference (CUI '26)*. ACM, 2026, pp. 1–22, <https://doi.org/10.1145/3816046.3816220>.

Khadka, Supriya, and Bijayan Bhattarai. “Gender Bias in Nepali-English Machine Translation: A Comparison of LLMs and Existing MT Systems”. *Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*. Association for Computational Linguistics, Aug. 2025, pp. 75–82, aclanthology.org/2025.gebnlp-1.6/.

Khadka, Supriya, Xiaorui Jiang, and Vasile Palade. “Automated Clinical Coding with Multiclass ICD Prediction”. *2025 International Conference on Machine Learning and Applications (ICMLA)*. IEEE, 2025, pp. 1295–300.

Khadka, Supriya, Ranju G.C., Prabin Paudel, Rahul Shah, and Basanta Joshi. “Nepali Text-to-Speech Synthesis using Tacotron2 for Melspectrogram Generation”. *Proc. 2nd Annual Meeting of the ELRA/ISCA SIG on Under-resourced Languages (SIGUL 2023)*. 2023, pp. 73–77, <https://doi.org/10.21437/SIGUL.2023-16>.

Peer-Reviewed Posters

Khadka, Supriya, and Sanchari Das. “Grant, Verify, Revoke: A User-Centric Pattern for Blockchain Compliance”. *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. 2026, pp. 1–6, <https://doi.org/10.1145/3772363.3798823>.

Khadka, Supriya, Dhiman Goswami, and Sanchari Das. “Poster: Privacy-Preserving Compliance Checks on Ethereum via Selective Disclosure”. *NDSS Symposium*, 2026. www.ndss-symposium.org/wp-content/uploads/ndss26-poster-32.pdf.

Manuscripts Under Review

Khadka, Supriya, Xiaorui Jiang, and Vasile Palade. “Data Quality in Clinical Coding: A Critical Analysis and Preliminary Study”. *medRxiv*, 2025. <https://doi.org/10.1101/2025.08.24.25334321>.

Sharma, Medha, **Supriya Khadka**, et al. “Evaluating Large Language Models’ Responses to Sexual, Maternal and Reproductive Health Queries in Nepali”. Manuscript under review at *Scientific Reports*. Medha Sharma and Supriya Khadka contributed equally., 2025.

Technical Skills

AI/Models	LLMs (GPT, Gemini, Llama, BLOOM), Transformers, NLP, ML
Programming	Python, JavaScript, PyTorch, Scikit-Learn
Tools & Tech	Git, Azure, MongoDB, Express, Node.js, SQL

Honors

Scholarships and Fellowships

- British Council Women in STEM Scholarship, 2024
- Ncell Excellence Scholarship - Yearwise Top Student, 2019 - 2023 [Press Releases [1](#), [2](#), [3](#), [4](#)]
- Fusemachines AI Fellowship, 2022
- 3rd Annual Nepal AI School - Full Scholarship, 2021
- Grace Hopper Conference Scholarship, 2021
- Women Leaders in Technology Fellowship, 2020

Awards

- Best Poster, 3rd Annual CSMM Conference, 2025
- GritFeat AI Health Hackathon, Runner Up, 2023
- Hack-A-Week Winner (Category - FinTech), 2022
- BPKHIS and GDHUB Hackathon for Healthcare, Runner Up 2022
- OpenIMIS Datathon, 2nd Runner Up, 2022

Service

Reviewer

- 3rd Annual CSMM Conference, 2025
- 7th International Workshop on Effective Engineering Education (IWEEE), 2022
- 6th International Workshop on Effective Engineering Education (IWEEE), 2021

Volunteering

- 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)

Research Posters and Presentations

- From Evidence to Codes: Making Automated Clinical Coding Explainable, **Poster**, *3rd Annual CSMM Conference, 2025*
- From Evidence to Codes: A Two-Stage Approach for Explainable Clinical Coding, **Oral**, *Researcher Capability and Development Conference (RCAD), 2025*
- Navigating Gender Biases in AI: Is it us or is it the data? A perspective on Nepal and beyond, **Oral**, *Women in Data Conference 2024* [[Press Release](#)]
- Shruti - Nepali Text-to-Speech Synthesis using Tacotron2 for Mel-Spectrogram Generation, **Poster and Oral**, *Young Women Leading the Way Showcase, 2023*

Student Activities

- Planning Coordinator, GitHub Field Day Nepal, 2023-24
- Managing Editor, The Zerone Magazine, 2022-23
- Event Manager, Hult Prize at IOE, Pulchowk Campus, 2021
- Technical Manager, Pulchowk Girls, 2020-22